



FUNDING PROPOSAL

STARBOARD MAIN ENGINE RESTORATION

Return Comanche's second 1944 Cleveland diesel-electric main to service so the ship keeps her own power — and her redundancy.

\$850,000

TOTAL REQUEST

Executive summary

Comanche is propelled by two Cleveland Diesel 12-cylinder diesel-electric main engines driving a single screw, original to her 1944 construction. The Foundation will fully survey, overhaul and return the starboard main engine and its generator to reliable service, restoring the two-engine redundancy the ship was designed with and preserving one of the last operating examples of this WWII powerplant.

Statement of need

Operating an 800-ton vessel on a single 80-year-old main engine leaves no margin: a failure underway is a safety event, not an inconvenience. The starboard main has been out of service pending overhaul; parts for the Cleveland 278-series are scarce and many must be machined or reverse-engineered. Restoring the engine is both a safety requirement for continued operation and a preservation act — these engines are as historic as the hull, and the volunteers who know them are aging out.

Project description

Scope: (1) engine survey — borescope, compression, crankshaft deflection and lube analysis; (2) disassembly, cleaning and non-destructive testing of crankshaft, rods, liners and heads; (3) parts — liners, pistons, rings, bearings, injectors, valves and gaskets, with an allowance for machining and reverse-engineering unavailable parts; (4) machine-shop work — line bore, head rebuild, crank polish; (5) main DC generator inspection, insulation testing, rewind or commutator work as found; (6) control, switchgear and cabling refurbishment; (7) cooling, lubrication and fuel-system renewal; (8) specialist marine-diesel labor with 278-series experience working alongside Foundation volunteers, who will be trained in the process; (9) dock trials and sea trials. Every step will be photographed and documented as a technical record for future stewards.

Goals & measurable objectives

- Starboard main engine and generator restored and passing dock and sea trials.
- Two-engine operation restored; single-point-of-failure risk removed.

- Complete technical documentation and parts inventory produced.
- At least four volunteers trained in Cleveland 278-series maintenance.
- Engine-room interpretation for visitors updated with the restoration story.

Timeline

Months 1–2	Survey, teardown, NDT; parts list and machining plan.
Months 2–6	Parts sourcing and machining; generator work in parallel.
Months 6–9	Reassembly, systems renewal, controls and cabling.
Month 10	Dock trials, sea trials, commissioning.
Months 11–12	Documentation, training completion, final report.

Budget

Engine survey: borescope, compression, crank deflection, oil analysis	\$16,000
Disassembly, cleaning and NDT of crank, rods, liners, heads	\$42,000
Parts: liners, pistons, rings, bearings, injectors, valves, gaskets	\$110,000
Machining and reverse-engineering allowance for unavailable parts	\$60,000
Machine-shop work: line bore, head rebuild, crank polish	\$58,000
Main DC generator: inspection, insulation test, rewind/commutator	\$75,000
Controls, switchgear and cabling refurbishment	\$36,000
Cooling, lubrication and fuel-system renewal	\$32,000
Specialist marine-diesel labor, ~900 hours	\$130,000
Rigging, lifting and engine-room access for major components	\$18,000
Trials: fuel, fluids, filters, surveyor attendance	\$14,000
Technical documentation and volunteer training program	\$9,000
Direct costs	\$600,000
Project management & administration (8%)	\$48,000
Contingency (30%)	\$194,400
Total project cost	\$842,400
Amount requested (rounded)	\$850,000

Budget lines are order-of-magnitude planning figures for a 133 ft / 295 GT steel vessel on Puget Sound, deliberately conservative so the funded scope can be delivered in full; they are replaced with vendor quotes at application. Unused contingency is returned or reallocated with funder approval. Partial funding is welcome and combinable with other sources.

In-kind and match. Foundation volunteer engineers and machinists (est. 1,200 hours), donated shop time sought from regional marine-diesel firms, historical drawings and manuals from the Foundation archive.

Evaluation

- Sea-trial report signed by the chief engineer and an independent marine surveyor.
- Parts and work log reconciled to budget.
- Documentation package archived; training roster on file.
- Engine hours logged in the first year of two-engine operation.

Funder fit

- National Maritime Heritage Grants
- Maritime Washington NHA
- Marine industry and engine-service sponsors
- Veterans and Coast Guard heritage foundations
- Private donors with maritime-engineering interest

Sustainability

With both mains serviceable the Foundation can alternate engines, reducing wear, and the trained volunteer cohort plus documentation package carry the knowledge forward. A machinery reserve is budgeted from operations after completion.

Organization overview

Tug Comanche Historical Rescue Foundation (EIN 39-5018917) is a Washington nonprofit based in Auburn, Washington, listed on the IRS Publication 78 data list as eligible to receive tax-deductible charitable contributions (deductibility code PC). The Foundation preserves the historic vessel Comanche — laid down in 1944 as the U.S. Navy ocean tug ATA-202, later USS Wampanoag and Coast Guard cutter WMEC-202 — and operates her as a living platform for maritime heritage, hands-on education, youth mentorship, restoration-skills exposure, veteran and mariner storytelling, Puget Sound stewardship, and community service. Comanche is one of the last operational WWII museum ships in the world, still moving under her own power with an all-volunteer crew. The organization is governed by a volunteer board of directors and delivers its programs through volunteers, maritime-trades mentors and community partners.

Vessel history & significance

Comanche is a rare surviving example of a WWII-era U.S. Navy auxiliary ocean tug. Completed in 1944 and commissioned as ATA-202, she was named USS Wampanoag in 1948 and later served as the U.S. Coast Guard cutter WMEC-202 before becoming known as Comanche. Her current U.S. Coast Guard Certificate of Documentation (Official Number 647250; IMO 8991786) records a steel-hulled, mechanically propelled vessel of 133.4 feet registered length, 33.1-foot breadth, 17.7-foot depth and 295 gross tons (ITC), hailing port Seattle, Washington. She is propelled by two Cleveland Diesel 12-cylinder diesel-electric main engines driving a single screw, and holds a world record as the last ship still steered by deck-tube chain rudder control.

Legal & operating boundary. Comanche does not hold a U.S. Coast Guard Certificate of Inspection for commercial passenger service. No grant funds, fees or donations are or will be used for paid carriage, charters, towing, freight or any underway commercial service. Public programs take place alongside the pier; cruises occur only in partnership with other nonprofit and community organizations, never as a charter, and the Foundation retains control of the vessel at all times.

Contact: Wesley Weaver, Secretary — info@tug202.org — tug202.org · The web version of this proposal, with the latest figures, is at tug202.org/grants/starboard-main-engine.